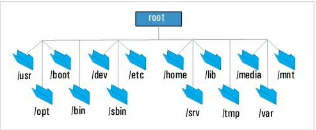

Figure 1: Linux file system hierarchy

of Linux. The *su* command is used to become a root or super user. Type the following command, and enter the root password to become a root or super user.

```
[bhargab@localhost~]$su
```

Handling files and directories

In Linux, 'Everything is a file'. This means that when we are dealing with normal text files or with device files, we interact with them through file operation related commands. Some operations on files are discussed below.

Creating a file: There are two commands to create a file: *touch* and *cat*. The *touch* command simply creates an empty file. Type the following command to create an empty document:

```
[bhargab@localhost~]$touch file1
```

cat is used to create and view a file. Type the following command to create a file:

```
[bhargab@localhost~]$cat>file1
```

To view a file type, use the command given below:

```
[bhargab@localhost~]$cat file1
```

Copying a file: The *cp* command is used to copy a file from one location to another, as shown below:

```
[bhargab@localhost~]$cp file1 /home/bhargab/Documents/
```

This command copies a file from the current working directory to */home/bhargab/Documents/*.

Removing a file: To remove a file, type the following command:

```
[bhargab@localhost~]$rm file1
```

Renaming and moving a file: The *mv* command is used to move and rename a file. To move a file from one location to another, use the following command:

```
[bhargab@localhost~]$mv file1 /home/bhargab/Document
```

The above command will move file1 to the *Document* directory under */home/bhargab/*.

To rename a file, file1 to file2, type the following command:

```
[bhargab@localhost~]$mv file1 file2
```

Listing files and directories: *ls* lists the contents (files and directories) of the current directory or specified directory. Type the following command to display the contents of the current directory:

```
[bhargab@localhost~]$ls
```

This command simply lists the file name and directory name. To list all files in your home directory, including the hidden files, type the following command:

```
[bhargab@localhost~]$ls -a
```

To view files in a 'long listing' format, type *ls* with the *–l* option, as follows:

```
[bhargab@localhost~]$ls -l
```

A portion of the output is shown below.

```
total 48

drwxr-xr-x. 2 bhargab bhargab 4096 Jan 25 21:32 Desktop
drwxr-xr-x. 2 bhargab bhargab 4096 Apr 24 16:33 Documents
drwxr-xr-x. 6 bhargab bhargab 4096 Jan 20 23:55 Downloads
-rw-rw-r--. 1 bhargab bhargab    1024 Apr 28 22:18 file1
-rw-rw-r--. 1 bhargab bhargab    1024 Apr 28 22:01 file2
-rw-rw-r--. 1 bhargab bhargab    1024 Apr 28 22:01 file3
drwxr-xr-x. 2 bhargab bhargab 4096 Dec 20 08:48 Music
drwxr-xr-x. 2 bhargab bhargab 4096 Dec 20 08:48 Pictures
drwxr-xr-x. 2 bhargab bhargab 4096 Dec 20 08:48 Public
drwxr-xr-x. 2 bhargab bhargab 4096 Dec 20 08:48 Videos
```

The total, 48, indicates that the total number of disk blocks occupied is 48. There are nine columns in each of the lines. Each column in the succession represents the following—permission, number of links, owner name, group name, size in bytes, date and time, and file name. The permission field consists of 10 sub-fields. The first field represents the type of file. The next three fields represent owner (u) permission. The fifth, sixth and seventh fields represent group (g) permissions. The last three fields represent other (o) permissions. 'w' represents write permission, 'x' represents execute permission and 'r' represents read permission.

Hard link and soft link

A link is a connection between a file name and actual data in the hard disk. There are two types of these -– the hard link and soft link.